

Project Plan

WBS, Gantt chart, communication plan and risk register

Project	Course	Case Study	Version
HaulTrack - 1,800-truck fleet telematics	B.Tech CSE - Software Engineering & Project Management	No. 19	1.0 (baseline)

1. Summary

Effort and schedule come from the Estimation Report: **239.1 person-months over about 17 months (74 weeks)**, an average team of about 14 FTE peaking near 20. The plan is built so that the union's concerns are managed as a **project risk with an owner, triggers and responses**, not as a communication afterthought.

2. Work Breakdown Structure

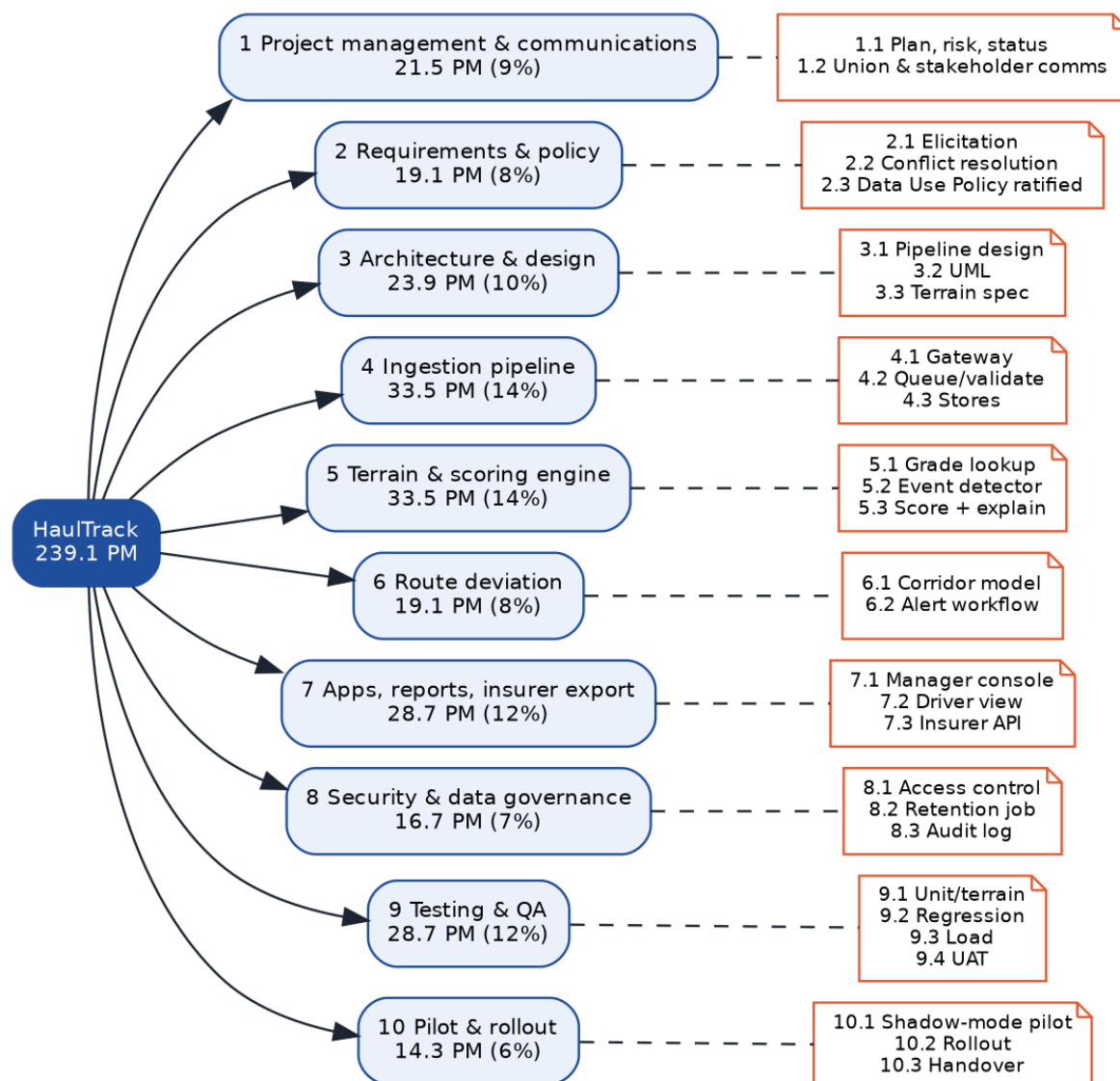


Figure 1 - WBS with person-month allocation (sums to the COCOMO estimate, 239.0 vs 239.1 from rounding).

WBS	Package	PM
1	Project management & communications	21.5
2	Requirements & policy	19.1
3	Architecture & design	23.9

WBS	Package	PM
4	Ingestion pipeline	33.5
5	Terrain & scoring engine	33.5
6	Route deviation	19.1
7	Apps, reports, insurer export	28.7
8	Security & data governance	16.7
9	Testing & QA	28.7
10	Pilot & rollout	14.3
	Total	239.0

3. Gantt chart

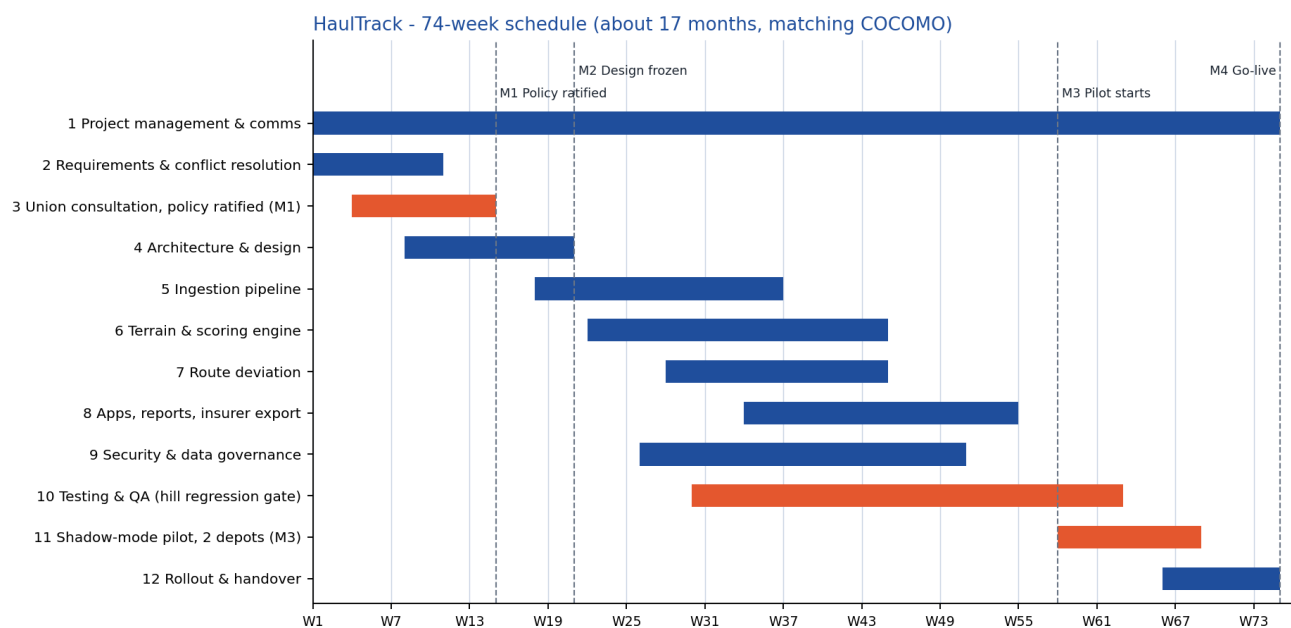


Figure 2 - Schedule. Coral bars are the stakeholder-critical items: policy ratification, testing with the hill-route gate, and the shadow pilot.

Milestone	Week	Gate
M1 Data Use Policy ratified	14	Union, Fleet Manager, Insurer liaison and Legal sign; scoring development builds against the signed policy
M2 Design frozen; estimate re-baselined	20	Design Pack approved; TN-1 to TN-7 accepted
M3 Shadow-mode pilot starts	57	Hill-route regression suite green; no score used for any decision
M4 Go-live	74	Pilot results reviewed by the Governance Board; TN-7 met

4. Stakeholder communication plan

Audience	What they need	Channel	Frequency	Owner	Feedback loop
Drivers' Union (rep + stewards)	What is measured, how terrain is handled, policy text, pilot results, any parameter change	Joint meeting + written summary; Governance Board seat	Fortnightly; Board monthly	Project Manager	Written responses logged; open issues carried to next meeting
Drivers	Plain-language guide, own-score preview, appeal route, what is NOT done with data	Depot briefings, FAQ sheet, self-view demo	Before pilot, at go-live, quarterly	Depot managers + union steward	Questions box; steward relays
Fleet Manager (sponsor)	Progress, risks, decisions needed, KPI results	Status report + review	Weekly	Project Manager	Decision log

Audience	What they need	Channel	Frequency	Owner	Feedback loop
Insurer liaison	Evidence format, metric definition, baseline, audit access	Evidence pack + call	Monthly in build; quarterly in service	Product owner	Agreed template; sign-off on definition
Legal / Data Protection	Retention, access, consent and lawful-basis review	Milestone reviews	At M1, M2, M3	Data governance lead	Review notes
IT Operations	Capacity, deployment, runbooks	Stand-up + handover sessions	Weekly	Technical lead	Issue tracker
Governance Board (Fleet Mgr, 2 union reps, insurer observer, DPO)	Metric versions, complaints, access reports, fairness statistics	Meeting + report	Monthly	Chair (rotating)	Minuted decisions

5. The union's concern as a project risk

Item	Detail
Risk statement	If the union believes the score can be used for arbitrary dismissal, it may refuse adoption or contest the system, blocking go-live and removing the pilot cohort.
Owner	Project Manager (with Fleet Manager as sponsor)
Early-warning indicators	Union does not sign the policy by M1; formal grievance; drivers disabling units; falling attendance at fortnightly meetings
Prevention	Policy written with the union and attached to the requirements; terrain fairness as measurable requirements; union seat on the Governance Board from week 4; shadow-mode pilot with no consequences; right to trigger an independent review
Response if triggered	Pause rollout; escalate to sponsor; extend the pilot; independent expert review of the score; policy amendment through the Board
Residual exposure	High-impact, medium probability: monitored at every status meeting

6. Risk register

ID	Risk	P	I	P x I	Mitigation / response
R1	Union rejects policy or refuses adoption	3	5	15	Section 5: early consultation, ratification gate M1, Board seat, independent review
R2	Hill-route unfairness recurs	3	5	15	TN-1 to TN-7, regression gate HR-01 to HR-05, pilot cohort analysis
R3	Score misused for discipline after go-live	3	5	15	Access control, no export path (PE-01), audit log, quarterly report to union, manager training
R7	Privacy breach or over-retention	2	5	10	Retention job (PE-04), minimisation, security review at M3
R6	Poor GPS or elevation data quality	3	3	9	Elevation accuracy test (ST-17), withhold score on gaps (ST-13)
R8	Effort or schedule overrun (COCOMO uncertainty)	3	3	9	Re-baseline at M2; phase-based staffing; no schedule compression
R11	Pressure for rankings (conflict with D5)	3	3	9	Decision log; Board approval needed for any visibility change
R12	Alert fatigue from false deviations	3	3	9	Suppression rules (RD-09 to RD-11); pilot tuning
R4	Insurer does not accept the metric or evidence	2	4	8	Agree KPI definition early; frozen v1.0; independent audit; capture baseline with same definition
R5	Telemetry volume or growth exceeds sizing	2	3	6	Partitioned queue; 3x peak headroom; load test
R9	Key staff loss	2	3	6	Pair work; documentation
R10	Previous vendor data unavailable for replay	3	2	6	Synthetic regression suite (HR-01 to HR-04) stands in

P and I are scored 1-5; exposure is P x I. Risks are reviewed weekly; anything at 15 or above is reported to the sponsor and the Governance Board.